

Homework 1

Please submit your answer to the Web Learning by next Thursday, March 12th.

1. Vector Operations

Consider two vectors

$$\mathbf{A} = 3\mathbf{i} - 2\mathbf{j}, \quad \mathbf{B} = -\mathbf{i} - 4\mathbf{j}.$$

Calculate:

(a) $\mathbf{A} + \mathbf{B}$ (b) $\mathbf{A} - \mathbf{B}$ (c) $|\mathbf{A} + \mathbf{B}|$ (d) $|\mathbf{A} - \mathbf{B}|$

(e) The directions of $\mathbf{A} + \mathbf{B}$ and $\mathbf{A} - \mathbf{B}$.

(f) If $\mathbf{A}(t) = 3t^2\mathbf{i} - 2t\mathbf{j}$ and $\mathbf{B}(t) = -t\mathbf{i} - 4t^2\mathbf{j}$, find $\frac{d}{dt}(\mathbf{A} \times \mathbf{B})$

2. Skydiver Motion

A skydiver jumps from a height of 1200 m. Initially falling without a parachute, the skydiver accelerates downward. Due to air resistance, the speed reaches a terminal velocity of 200 km/h, after which the descent continues at constant speed.

When the skydiver is 50 m above the ground, the parachute is deployed, quickly reducing the speed to 18 km/h. The final descent to the ground is at this constant speed.

If the average acceleration during the initial phase is $g/2$, find the total time the skydiver spends in the air.

3. Projectile Over a Wall

A person can throw a stone with a maximum initial speed of 25 m/s.

Can he throw the stone over a wall that is at a horizontal distance $L = 50$ m and has a height $h = 13$ m?

Within this horizontal distance, what is the maximum height the stone can reach?

4. Projectile on an Incline

A projectile is fired up an incline (incline angle ϕ) with an initial speed v_i at an angle θ_i with respect to the horizontal ($\theta_i > \phi$), as shown in Figure P4.47.

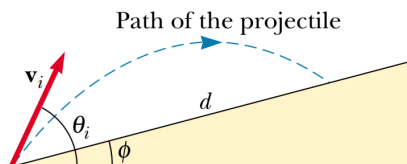


Figure P4.47

(a) Show that the projectile travels a distance d up the incline, where

$$d = \frac{2v_i^2 \cos \theta_i \sin(\theta_i - \phi)}{g \cos^2 \phi}.$$

(b) For what value of θ_i is d a maximum? What is the maximum value of d ?

5. Train Rounding a Curve

A train slows down as it rounds a sharp horizontal curve, decreasing its speed from 90.0 km/h to 50.0 km/h in the 15.0 s it takes to round the curve. The radius of curvature is 150 m.

Compute the acceleration of the train at the moment it reaches 50.0 km/h. Assume the train slows at a uniform rate during the 15.0 s interval.

6. Elevator and Free Fall

An elevator descends with a constant acceleration of 1.2 m/s^2 . A passenger releases a small ball from a height of 1.5 m above the elevator floor, 0.5 s after the elevator begins descending.

Find:

- (a) The time required for the ball to reach the floor.
- (b) The distance the ball falls relative to the ground.